MIL-I-6726B(USAF)
16 Nov 1966
Superseding
MIL-I-6726A(ASG)
10 June 1954

MILITARY SPECIFICATION

INCLINOMETER, AIRCRAFT

1. SCOPE

1.1 This specification covers one type of 7/16 inch diameter ball-style inclinometer.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-D-1000	Drawings, Engineering and Associated Lists
MIL-L-5020	Liquid, Compass, Aircraft
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification
MIL L 25142	Luminescent Material, Fluorescent

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-143	Specifications and Standards, Order of Preference for the Selection of
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MS33586	Metals, Definition of Dissimilar

Air Force-Navy Aeronautical

AN5744	Inclinometer-Aircraft, Ball Style

FSC 6610

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(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Qualification. The inclinometer furnished under this specification shall be a product which has been tested, and passed the qualification tests specified herein, and has been listed on or approved for listing on the applicable qualified products list.

3.2 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.3 Materials

3.3.1 Protective treatment. When materials are used in the construction of the inclinometer that are subject to corrosion in salt air or other atmospheric conditions likely to be encountered during service usage, they shall be protected against corrosion in a manner which will in no way prevent compliance with the requirements of this specification. The use of any protective coating that will crack, chip or scale with age or extremes of atmospheric conditions shall be avoided.

3.3.2 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MS33586.

3.4 Design and construction.

3.4.1 General. The inclinometer shall be constructed so that no parts will work loose in service. It shall be built to withstand the strains, jars, vibrations and other conditions incident to shipping, storage, installation and service.

3.4.2 The inclinometer shall conform to AN5744.

3.4.3 Ball. The ball shall be made of black glass or similar material. It shall be highly polished and accurately spherical. The diameter of the ball shall be 7/16 $\pm$ 1/64 inch.

3.4.4 Tube. The glass tube shall be made of clear, annealed glass tubing, free from defects which may interfere in any way with normal reading of the inclinometer. The wall thickness shall not be less than 3/64 inch. The inside of the tube shall be smooth and uniform in order to insure that the ball will roll

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smoothly. The tube shall be firmly secured in the frame by means of a filler. An expansion chamber shall be provided at each end of the tube.

3.4.5 Damping fluid. The damping fluid shall conform to MIL-L-5020 and shall be sufficiently colorless not to interfere with satisfactory silhouette illumination of the ball by luminescent material.

3.4.6 Frame. The frame shall be fabricated of aluminum alloy.

3.4.7 Fluorescent material. The fluorescent material shall conform to MIL-L-25142.

3.5 Installation.

3.5.1 Mounting screws. The contractor shall furnish sufficient mounting screws for installing the inclinometer. They shall be No 6-32 round head, brass machine screws, one inch long, durable black finish and shall be fitted with lock washers and brass nuts.

3.5.2 Envelope. An envelope, furnished by the contractor, containing the mounting screws, and having the following information printed on the face shall be provided:

IMPORTANT
THIS ENVELOPE CONTAINS MOUNTING SCREWS

3.6 Performance. The inclinometer shall be capable of passing the performance tests specified in Section 4.

3.7 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The drawing number requirements of MIL-D-1000 shall govern changes in the manufacturer's part numbers.

3.8 Operation markings. The markings on the inclinometer shall be in accordance with Table I.

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TABLE I

INCLINOMETER MARKINGS			
Markings	Height or Length	Width of Line or Graduations (inch ± .005)	Material or Finish
Numerals	0.150 $^{+0.010}_{-0.000}$	0.030	
5- and 15-degree lines	0.125 ± 0.010	0.030	Fluorescent Luminescent Material
Intermediate degree lines	0.093 ± 0.010	0.030	
Triangles at 0, 10, and 20	0.093 ± 0.010 inch sides equilateral triangles		
Rear portion of glass tube			
Exterior metallic surfaces			Durable dull black

3.8.1 All markings shall be durable to withstand usage encountered in service.

3.9 Identification of product. The inclinometer shall be marked for identification in accordance with MIL-STD-130.

3.10 Weight. The weight of the inclinometer shall not exceed 0.3 pound.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to the prescribed requirements.

4.2 Classification of tests. The inspection and testing of inclinometers shall be classified as follows:

a. Qualification tests

b. Acceptance tests

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4.3 Qualification tests. The qualification tests shall consist of all the tests of this specification.

4.3.1 Qualification test samples. Qualification tests shall be performed on two samples. These samples shall be identified as qualification samples and forwarded to the activity designated in the letter of authorization (see 6.3).

4.4 Acceptance tests. Acceptance tests shall consist of individual tests and sampling tests.

4.4.1 Individual tests. Each inclinometer submitted for acceptance under contract shall be subjected to the following tests as described in 4.6:

a. Examination of product

b. Filling

c. Scale error

d. Ball visibility

4.4.2 Sampling tests. In addition to the individual tests, three inclinometers shall be selected at random from each lot of 100, or fraction thereof, subjected to the following tests, as described in 4.6:

a. Damping, room temperature

b. Damping, low temperature

c. Vibration

d. Leak

4.4.2.1 In addition to the above, two of the three samples shall be subjected to the following tests, one sample for each test:

a. Salt spray

b. Mildew resistance

4.4.2.2 Rejection and retest. When one or more inclinometers from a lot fail to pass one or more of the sampling tests, acceptance of the lot will be withheld until the extent and cause of failure are determined. After corrections have been made, six additional inclinometers shall be selected at random from the lot and subjected to the sampling tests. The lot shall be accepted if no defects are found. If any defect is found in the second sample, disposition of the lot will be determined by the procuring activity.

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4.4.2.3 Individual tests may continue. For production reasons, individual tests may be continued pending the investigation of a sampling test failure. Final acceptance of the lot will not be made, however, until it is determined that the lot meets all the requirements of the specification.

4.4.2.4. Defects in items already accepted. The investigation of a sampling test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.5 Test conditions.

4.5.1 Atmospheric conditions. Unless otherwise specified, all tests are to be conducted at an atmospheric pressure of approximately 29.92 inches of mercury and at a temperature of approximately 77°F. When tests are conducted at conditions substantially different from the above values, proper allowances shall be made for the differences which could change instrument readings or inclinometer performance.

4.5.2 Tapping. Unless otherwise specified, the inclinometer shall be gently tapped or vibrated before a test reading is taken.

4.5.3 Attitude. Unless otherwise specified, the inclinometer shall be tested in normal operating position.

4.5.4 Vibration. A vibration stand shall be used which will vibrate at any desired frequency between 500 and 3,000 cycles per minute and shall subject the inclinometer to vibration such that a point on the frame will describe, in a plane inclined 45 degrees to the horizontal plane, a circle with a diameter of not less than 0.018 nor more than 0.020 inch.

4.6 Test methods.

4.6.1 Examination of product. Each inclinometer shall be examined to determine conformance to the specification with respect to materials, dimensions, weight, marking and general requirements.

4.6.2 Filling. The inclinometer shall be tipped in such a manner that all the air in the tube is trapped in one end of the tube. Then with the back of the inclinometer in a vertical plane and with the line joining the center of the mounting holes inclined to the horizontal at an angle of 20 degrees, with the end of the tube containing the bubble downward, no part of the air bubble shall be visible when the instrument is viewed from a position 12 inches directly in front of the zero mark of the inclinometer

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4.6.3 Scale error. The inclinometer shall be tested for scale errors at the angles of inclination specified in Table II. All scale errors shall be determined with the back of the instrument in a vertical plane and with gentle tapping or vibration of the instrument before a test reading is taken. The error at the zero point of the scale shall be determined with the line joining the centers of the mounting holes inclined to the horizontal at the specified angles. The ball shall roll smoothly throughout the length of the tube included between the 20-degree marks when the inclinometer is slowly inclined to the horizontal, with the back of the instrument in a vertical plane, and with gentle tapping or vibration of the inclinometer. The ball shall not stick at either end of the tube.

TABLE II

Scale Errors

Inclination-degrees	Tolerance-degrees
0	0.5
5	1.0
10	1.5
15	2.0
20	2.0

4.6.4 Ball visibility. With the inclinometer tipped in such a manner that the ball is at rest at either end of the tube, not less than half of the ball shall be visible when viewed from a position 12 inches directly in front of the zero mark of the inclinometer.

4.6.5 Damping, room temperature. With the back of the inclinometer in a vertical plane and with the line joining the centers of the mounting holes inclined to the horizontal at an angle of 24 degrees, the inclinometer shall be tipped suddenly through a vertical angle of 48 degrees in order to insure that the ball will roll to the opposite end of the tube. The time of roll of the ball from the zero mark to the 20-degree mark of the inclinometer shall not be less than 1/2 second.

4.6.6 Damping, low temperature. The inclinometer shall be tested as specified in 4.6.5, except that it shall be at a temperature of minus 130 ± 2°F. The time of roll of the ball shall not exceed 10 seconds.

4.6.7 Vibration. The inclinometer shall be mounted on a suitable vibration stand with the line joining the centers of the mounting holes in a horizontal plane, and with the back of the inclinometer in a vertical plane. The inclinometer shall then be subjected to

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vibration as specified in 4.5.4. While the inclinometer is being vibrated, the instrument reading shall not differ by more than one degree at any time from the instrument reading obtained while the inclinometer was being lightly tapped.

4.6.8 Leak. The size of the air bubble in the inclinometer shall be determined with the instrument at room temperature. The inclinometer shall then be maintained at a temperature of 160 ± 3°F for a period of not less than one hour. The size of the air bubble shall then be determined with the inclinometer at room temperature. There shall be no appreciable change in the size of the air bubble before and after the inclinometer has been subjected to the temperature specified.

4.6.9 Salt spray. The salt spray test shall be conducted in accordance with MIL-E-5272. The duration of the test shall be 50 hours. After the test, the inclinometer shall be examined. There shall be no evidence of corrosion or damage that would affect subsequent operation.

4.6.10 Mildew resistance. The mildew resistance test shall be conducted in accordance with Procedure I of MIL-E-5272. After the test, the inclinometer shall meet the test requirements of 4.6.3. There shall be no deterioration which would adversely affect subsequent operation.

5. PREPARATION FOR DELIVERY

5.1 Packaging and packing. The inclinometer shall be preserved and packaged in accordance with Level A of MIL-STD-794, unless otherwise specified by the procuring activity.

5.1.1 Packing for shipment shall be in accordance with Level B of MIL-STD-794 unless otherwise specified by the procuring activity.

5.2 Marking. In addition to any special marking required by the contract or order, unit packages, intermediate packages and shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

6. NOTES

6.1 Intended use. The inclinometer covered by this specification is intended for use on aircraft to indicate lateral attitude of the aircraft to the apparent vertical.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification

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b. Level of packaging and packing required.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved forinclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is Oklahoma City Air Materiel Area (OCNE), Tinker AFB, Oklahoma, and information pertaining to qualification of products may be obtained from that activity.

CUSTODIAN:
Air Force - 71

PREPARING ACTIVITY:
Air Force - 71

REVIEWER:
Air Force - 71

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor involved in the use of the specification in procurement of products for ultimate use by the Department of Defense This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost Comments and the return of this form will be appreciated Fold on lines on reverse side staple in corner and send to preparing activity

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF 'YES' IN WHAT WAY?

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification If there are additional papers attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863 WHICH IS OBSOLETE Air Force OC, 24 Dec 64-500

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